EPSR-test-ex98 revlorch: Q-space Lorch broadening

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the `revlorch` option of the `epsr` engine of EPSR [1,2], which broadens the r -> Q transform of the simulated g(r) with a Lorch-type window [3] whose width grows linearly with Q, to mimic resolution or disorder broadening in the simulated structure factors [4].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (reading `lorchqwidth`, `lorchqwidthq`)
- `epsr_standalone_rev3.f`: `inita` (transform matrix `foqols`: `bwidth` = `rr*lorchlim*(lorchqwidth + lorchqwidthq*qq)`, `lorchlim` = `4.4934/rmaxgr`), function `pofr`

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::setup` (`revlorch`)
- `structure.rs`: `inita` (`foqols`), `pofr`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [5] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [6]. For this case `revlorch` `0.05` `0.01` multiplies each element of the transform matrix by the single-precision window `pofr(bwidth)`, so every simulated S(Q) (`.q01`, `.u01`, `.x01`, ...) is broadened relative to the base case. With `rmaxgr` the maximum radius of the g(r), `lorchlim` = `4.4934/rmaxgr` sets the scale, multiplied by the Q-dependent factor (0.05 + 0.01·Q).

Inputs

Only the `.EPSR.inp` differs from EPSR-test, edited by `scripts/make_inp.sh` (via `scripts/setkw.py`): `revlorch` `0.05` `0.01`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Because the broadened matrix enters every simulated structure factor and hence the EP update, all S(Q) outputs and the subsequent potentials and configurations differ from the base case and are identical between Fortran and port in each pass. The window function `pofr` is evaluated in single precision as in the Fortran, so the transform matrix `foqols` agrees to the last bit, which the identical `.q01`, `.u01` and `.x01` files confirm.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] E. Lorch, "Neutron diffraction by germania, silica and radiation-damaged silica glasses", J. Phys. C: Solid State Phys. 2, 229-237 (1969).
[4] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[5] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[6] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[7] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.